Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2001-I-Q03] | Part A (Q1-Q5)

Answer: 500

1. The degree-2001 terms cancel. The resulting degree-2000 polynomial has leading coefficient $2001/2$ and next coefficient $-\binom{2001}{2}/4$.

2. Vieta gives root sum $\binom{2001}{2}/(2 \cdot 2001) = 2000/4 = 500$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2013-I-Q05] | Part A (Q1-Q5)

Answer: 098

1. The equation is $9x^3 = (x+1)^3$. Taking real cube roots gives $\sqrt[3]{9}\,x = x+1$, hence $x = 1/(\sqrt[3]{9}-1)$.

2. Use $(u-1)(u^2+u+1) = u^3-1$ to rationalize: $x = (\sqrt[3]{81}+\sqrt[3]{9}+1)/8$. The linear equation after cube roots also proves uniqueness.

3. The requested sum is $81+9+8 = 98$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2014-II-Q05] | Part A (Q1-Q5)

Answer: 420

1. Let the third root of p be $t = -r - s$. The other cubic has third root $t - 1$. Equality of their pairwise root-product sums gives $t = 4r - 3s + 13$.

2. The constant-term difference is $rst - (r + 4)(s - 3)(t - 1) = 240$. Using the preceding linear relation reduces this to $(r - s)^2 + 7(r - s) - 8 = 0$. Thus $r - s = 1$ or -8.

3. Combining with $t = -r - s$ gives the respective root triples $(-5, -6, 11)$ and $(1, 9, -10)$. Their constant terms are -330 and 90. Each shifted triple verifies the required second cubic.

4. The absolute-value sum is 420.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2018-I-Q01] | Part A (Q1-Q5)

Answer: 600

1. Write the factors $(x+r)(x+s)$ with $0 \le r \le s$. Since their sum is positive and product nonnegative, this covers every integer-root possibility. The condition is $r+s=a$, with $b=rs$ uniquely determined.

2. For each a, there are $\lfloor a/2 \rfloor + 1$ choices. Pair $a=2k-1, 2k$ for $k=1,\ldots,50$; their contribution is $k+(k+1)$.

3. The total is $2(50 \cdot 51/2) + 50 = 2600$, leaving 600.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2021-II-Q04] | Part A (Q1-Q5)

Answer: 330

1. Both also have root $m - i\sqrt{n}$. The first cubic has zero quadratic coefficient, so $-20 + 2m = 0$ and $m = 10$.

2. The second has zero linear coefficient, so the pairwise root products sum to zero: $m^2 + n - 42m = 0$. Hence $n = 320$, and $m + n = 330$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2022-I-Q01] | Part A (Q1-Q5)

Answer: 116

1. Add the polynomials first. Their quadratic terms cancel, so their sum R is linear, with $R(16) = 108$ and $R(20) = 106$.

2. The slope is $-1/2$. Moving from $x = 16$ back to 0 increases R by 8, giving $R(0) = 116$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2003-II-Q09] | Part B (Q6-Q10)

Answer: 006

1. Division gives $P = (x^2+1)Q + (x^2 - x + 1)$. Therefore at every root, $P(z_i) = z_i^2 - z_i + 1$.

2. Vieta gives $\sum z_i = 1$, $\sum_{i<j} z_i z_j = -1$, so $\sum z_i^2 = 3$. The requested sum is $3 - 1 + 4 = 6$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2004-I-Q07] | Part B (Q6-Q10)

Answer: 588

1. With $u_j = (-1)^j j$, the coefficient is $\sum_{i<j} u_i u_j = [(\sum u_j)^2 - \sum u_j^2]/2$.

2. The alternating sum is -8, and $\sum_{j=1}^{15} j^2 = 1240$. Thus $C = (64 - 1240)/2 = -588$, so $|C| = 588$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2005-II-Q07] | Part B (Q6-Q10)

Answer: 125

1. Put $y = 5^{1/16}$. Multiplying the denominator by $y - 1$ repeatedly uses difference of squares, producing $(y-1)(y+1)(y^2+1)(y^4+1)(y^8+1) = y^{16} - 1 = 4$.

2. Therefore $x = y - 1$, so $(x+1)^{48} = y^{48} = 5^3 = 125$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2007-I-Q08] | Part B (Q6-Q10)

Answer: 030

1. The two quadratics must share a root, since otherwise their degree-four product divides a cubic. Twice the first minus the second is $-15x - 3k$, so a common root is $x = -k/5$.

2. Substitution into the first gives $-4k^2 + 120k = 0$, hence $k = 0$ or 30. For 30, the factors are $(x + 6)(x - 5)$ and $(x + 6)(2x + 5)$, which indeed divide their cubic least common multiple.

3. The maximum is 30.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2008-II-Q07] | Part B (Q6-Q10)

Answer: 753

1. Vieta gives $r+s+t=0$ and $rst=-251$. Each pair sum is the negative remaining root, so the requested sum is $-(r^3+s^3+t^3)$.

2. When the root sum is zero, the cube-sum identity yields $r^3+s^3+t^3=3rst=-753$. The answer is 753.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2010-I-Q06] | Part B (Q6-Q10)

Answer: 406

1. The bounds are $(x-1)^2+1$ and $2(x-1)^2+1$, so $P(1)=1$. The nonnegative quadratic $P(x)-[(x-1)^2+1]$ vanishes at 1 and must have a double root there.

2. Consequently $P(x)=a(x-1)^2+1$. Its value at 11 gives $a=9/5$, within the required interval $[1,2]$.

3. Therefore $P(16)=(9/5)225+1=406$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2010-II-Q06] | Part B (Q6-Q10)

Answer: 008

1. If an integer root r exists, $n = r^3 + 63/r$, with r a positive divisor of 63. For $r = 1, 3$ this gives $64, 48$; for $r \geq 7$ it exceeds 48. Negative r give negative n.

2. Otherwise monicity permits a factorization $(x^2 + px + q)(x^2 - px + r)$ with integers. Matching coefficients gives $qr = 63, q + r = p^2, n = p(q - r)$. Negative q, r cannot have square sum; positive factor pairs have sums $64, 24, 16$.

3. Only pairs $(1, 63), (7, 9)$ have square sums. Their positive possible n are $496, 8$.

4. The minimum is 8, attained by $(x^2 + 4x + 9)(x^2 - 4x + 7)$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2014-I-Q06] | Part B (Q6-Q10)

Answer: 036

1. The expanded quadratics have root sums $2h$ and root products $2013/3 = 671 = 11 \cdot 61$ and $2014/2 = 1007 = 19 \cdot 53$.

2. Positive factor pairs give possible sums $72, 672$ for the first and $72, 1008$ for the second. The only common sum is seventy-two.

3. Therefore $h = 36$, attained by roots $11, 61$ and $19, 53$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2014-I-Q09] | Part B (Q6-Q10)

Answer: 002

1. Put $a = \sqrt{2014}$. The cubic factors as $(ax - 1)(x^2 - 2ax - 2)$.

2. The quadratic roots are $a \pm \sqrt{a^2 + 2}$, one negative and one greater than a. Thus the middle root is $1/a$, while the other two sum to $2a$.

3. The product requested is 2.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2015-I-Q10] | Part B (Q6-Q10)

Answer: 072

1. Each cubic $f-12, f+12$ has at most three distinct roots, so the six inputs split into two root triples. Since the cubics differ only by a constant, the two triples have equal sums and equal pairwise-product sums.

2. The total root sum is 24. The only three-element subsets with sum 12 are $\{1,5,6\}$ and $\{2,3,7\}$, complementary and both having pairwise-product sum 41.

3. Change the sign of f if needed. Then $f=c(x-2)(x-3)(x-7)+12=c(x-1)(x-5)(x-6)-12$. At $x=0$, $-42c+12=-30c-12$, so $c=2$ and $f(0)=-72$. The requested absolute value is 72.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2015-II-Q06] | Part B (Q6-Q10)

Answer: 440

1. If the roots are $r \le s \le t$, Vieta gives $r+s+t=a$ and $rs+rt+st=(a^2-81)/2$. Subtraction yields $r^2+s^2+t^2=81$.

2. As $r \le \sqrt{27} < 6$, test $r=1,2,3,4,5$ and $r \le s \le \sqrt{(81-r^2)/2}$. The only square remainders give triples $(1,4,8),(3,6,6),(4,4,7)$.

3. The root sums are respectively $13, 15, 15$. The stated ambiguity therefore means $a=15$. Since $c=2rst$, the two values are 216 and 224, with sum 440.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2016-II-Q06] | Part B (Q6-Q10)

Answer: 275

1. All substituted exponents are odd, so replacing x by $-x$ makes every factor have nonnegative coefficients. Thus the coefficient $(-1)^j a_j$ of $Q(-x)$ is nonnegative, and equals $|a_j|$.

2. Summing the coefficients evaluates at $x = 1$: $\sum |a_j| = Q(-1) = P(-1)^5 = (3/2)^5 = 243/32$. Therefore $m + n = 275$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2018-II-Q06] | Part B (Q6-Q10)

Answer: 037

1. The polynomial factors as $(x-1)(x+2)[x^2+(2a-1)x+1]$. Only the quadratic can have nonreal roots.

2. Its discriminant is nonnegative exactly when $|2a-1| \geq 2$, or $a \leq -1/2$ or $a \geq 3/2$. This excludes an interval of length two from a sample interval of length thirty-eight.

3. The probability is $36/38 = 18/19$, giving 37.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2019-I-Q08] | Part B (Q6-Q10)

Answer: 067

1. Let $u = \sin^2 x, v = \cos^2 x, p = uv$, so $u + v = 1$ and $0 \le p \le 1/4$. The fifth power sum is $u^5 + v^5 = 1 - 5p + 5p^2$, giving $36p^2 - 36p + 5 = 0$.

2. The roots are $1/6, 5/6$; the bound selects $p = 1/6$. The sixth power sum is $1 - 6p + 9p^2 - 2p^3 = 13/54$.

3. The answer is 67.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2019-I-Q10] | Part B (Q6-Q10)

Answer: 352

1. Vieta on roots repeated three times gives $T = \sum z_j = -20/3$. Let $U = \sum_{j<k} z_j z_k$. The next coefficient counts three same-root-copy pairs and nine cross-root-copy pairs, so $19 = 3\sum z_j^2 + 9U$.

2. Substitute $\sum z_j^2 = T^2 - 2U$, obtaining $19 = 400/3 + 3U$. Thus $U = -343/9$, absolute value $343/9$, and the answer is 352.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2023-I-Q09] | Part B (Q6-Q10)

Answer: 738

1. Factor $p(x) - p(2) = (x-2)[x^2 + (a+2)x + 2a + b + 4]$. The constant c is unrestricted, supplying a factor of 41.

2. A monic integer quadratic with one integer root has two integer roots. Thus the bracket must have roots $2, k$ with $k \neq 2$, or repeated root $k \neq 2$.

3. The first case gives $a = -k - 4, b = 4k + 4$ and $-6 \leq k \leq 4$, excluding 2: ten choices. The second gives $a = -2k - 2, b = k^2 + 4k$ and $-6 \leq k \leq 2$, again excluding 2: eight choices.

4. These cases are disjoint, and every listed coefficient is in range. The total is $(10 + 8)41 = 738$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2000-II-Q13] | Part C (Q11-Q15)

Answer: 200

1. Use the cube difference $(10x^2)^3 - 1$ to factor the polynomial as $(100x^4 + 10x^2 + 1)(20x^2 + x - 2)$.

2. The first factor is positive on the real line. The quadratic roots are $(-1 \pm \sqrt{161})/40$, so the root with the positive radical has $(m, n, r) = (-1, 161, 40)$ and the sum is 200.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2008-I-Q13] | Part C (Q11-Q15)

Answer: 040

1. The axial and $(1, \pm 1)$ zeros reduce the polynomial to $u(x - x^3) + v(y - y^3) + wxy(1 - x)$. The zero at $(2, 2)$ imposes $w = -3(u + v)/2$.

2. A universal zero must annihilate both $x - x^3 - 3xy(1 - x)/2$ and $y - y^3 - 3xy(1 - x)/2$. Positive nonintegral x makes the first condition $y = 2(x + 1)/3$.

3. Substitution into the second yields $19x^2 - 43x + 10 = 0$, with roots $2, 5/19$. The new nonintegral point is $(5/19, 16/19)$, so $a + b + c = 40$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2013-II-Q12] | Part C (Q11-Q15)

Answer: 540

1. If all roots are real, they are a multiset of three from $\{-20, -13, 13, 20\}$, giving $\binom{6}{3} = 20$ polynomials.

2. Otherwise the unique real root is one of these four integers, say k, and the remaining roots are a nonreal conjugate pair of squared modulus $R^2 = 169$ or 400. Their quadratic factor is $z^2 - tz + R^2$, where integer coefficients require exactly that t be an integer.

3. Nonreality requires $|t| < 2R$. This allows 51 values when $R = 13$ and 79 when $R = 20$. Every choice gives a distinct integer cubic.

4. The total is $20 + 4(51 + 79) = 540$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2016-I-Q11] | Part C (Q11-Q15)

Answer: 109

1. Substitute $x = 1, 0, -2$ to obtain roots $1, 0, -1$. Thus $P = x(x-1)(x+1)L(x)$ for a polynomial L.

2. The functional identity reduces, as a polynomial identity, to $L(x+1) = L(x)$. A nonconstant polynomial has a nonzero finite difference of degree one lower, so L must be constant $c \neq 0$.

3. The normalization gives $(6c)^2 = 24c$, hence $c = 2/3$. Then $P(7/2) = (2/3)(7/2)(5/2)(9/2) = 105/4$, giving 109.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2020-I-Q11] | Part C (Q11-Q15)

Answer: 510

1. The inputs $u = f(2)$ and $v = f(4)$ coincide precisely when $a = -6$. Then every b, c works by setting $d = -u^2 - cu$, giving 441 triples.

2. Otherwise u, v are distinct roots, so $c = -u - v = -20 - 6a - 2b$. Thus $-15 \leq 3a + b \leq -5$, with $b \in [-10, 10]$.

3. For $a = -8, -7, -5, -4, -3, -2, -1, 0, 1$ the counts of b are $2, 5, 11, 11, 11, 11, 9, 6, 3$; all other allowed $a \neq -6$ contribute zero. Their sum is 69, and the total is 510.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2020-I-Q14] | Part C (Q11-Q15)

Answer: 085

1. Write $P = x^2 - sx + q$. The four roots split into two pairs, each satisfying $P(x) = r$ for one root r of P; each pair sums to s.

2. If $3, 4$ are paired, then $s = 7$ and $a + b = 7$, giving 49. This case exists: take $q = 12 + r$ with $r^2 - 6r + 12 = 0$; the other pair is distinct from $3, 4$ and from itself.

3. If they are in different pairs, the other roots are $s - 3, s - 4$. Since $P(3), P(4)$ are the roots of P, their sum gives $q = 4s - 25/2$. Thus $P(3) = s - 7/2, P(4) = 7/2$, and their product condition gives $(7/2)(s - 7/2) = 4s - 25/2$.

4. Solve $s = 1/2$, obtaining $a + b = -6$ and 36. All four roots $3, 4, -5/2, -7/2$ are distinct, so the final sum is $49 + 36 = 85$.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2020-II-Q11] | Part C (Q11-Q15)

Answer: 071

1. Name the shared roots so that $P+Q=2(x-a)(x-b)$, $P+R=2(x-a)(x-c)$, $Q+R=2(x-b)(x-c)$. Combining these expressions recovers $2P$, whose linear coefficient yields $a=3/2$.

2. The constant term of $P+Q$ gives $2ab=-5$, hence $b=-5/3$. If $s=R(0)$, the other constants give $2ac=s-7$ and $2bc=s+2$.

3. Subtract to get $2c(b-a)=9$, so $c=-27/19$. Then $s=2bc-2=52/19$; the three roots are distinct and the answer is 71.

Source / 来源

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2024-II-Q11] | Part C (Q11-Q15)

Answer: 601

1. Set $q = ab + bc + ca, p = abc$. The degree-three sum is $300q - 3p$, so the condition is $100q - p = 2000000$. Expanding shows this is equivalent to $(100 - a)(100 - b)(100 - c) = 0$.

2. For each specified variable equal to 100, the other two sum to 200, giving 201 choices. Any overlap forces $(100, 100, 100)$, counted three times.

3. Correct the overcount: $3 \cdot 201 - 2 = 601$.

Source / 来源